

Training Log Interpreter

This notebook parses and visualizes training logs from the Int2Int model training. It extracts model parameters, task information, and plots metrics over epochs.

Configuration

Specify the path to your training log file:

Parse Training Log

Parsed 103 parameters and 201 epoch logs

Report Model Configuration and Training Parameters

```
=====
MODEL CONFIGURATION AND TRAINING PARAMETERS
=====
```

```
- EXPERIMENT INFO
```

```
Experiment Name: mu
Experiment ID: 1
Task Type: MU
Operation: data
```

```
- DATA
```

```
Training Data: /mnt/c/Users/ziwen/clair/mobius_case_study/input/input_dir_interCRT_
sqfree_natural/mu_interCRT_sqfree_natural.txt.train
Eval Data: /mnt/c/Users/ziwen/clair/mobius_case_study/input/input_dir_interCRT_sqfr
ee_natural/mu_interCRT_sqfree_natural.txt.test
Data Types: int[201]:range(-1,2)
Base: 1000
Modulus: 67
```

```
- MODEL ARCHITECTURE
```

```
Architecture: encoder_decoder
Encoder Layers: 4
Decoder Layers: 4
Encoder Embedding Dim: 256
Decoder Embedding Dim: 256
Encoder Heads: 8
Decoder Heads: 8
Dropout: 0.1
Attention Dropout: 0.1
```

```
- TRAINING PARAMETERS
```

```
Optimizer: adam_inverse_sqrt,lr=0.00025
Batch Size: 128
Eval Batch Size: 128
Epoch Size: 50000
Max Epochs: 201
Eval Size: 10000
Gradient Clipping: 5
Max Length: 512
Max Output Length: 512
```

```
- OTHER SETTINGS
```

```
FP16: False
CPU Mode: False
Multi-GPU: False
Num Workers: 0
=====
```

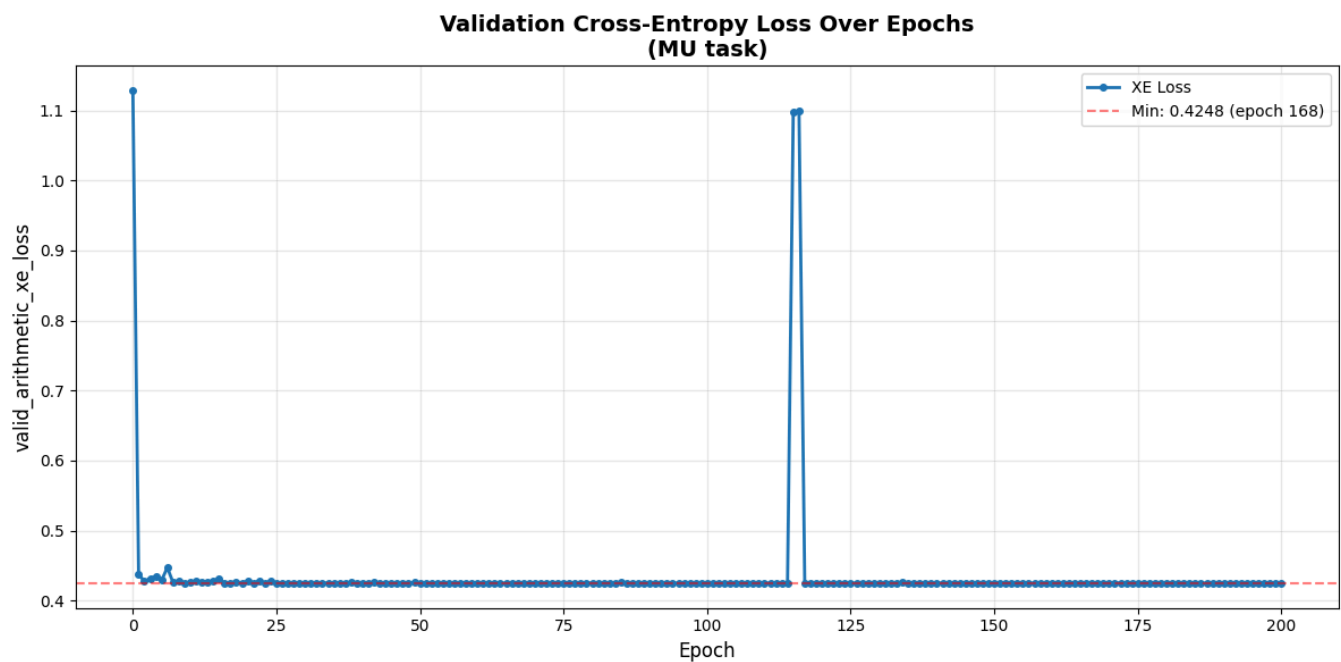
Convert Metrics to DataFrame

Total epochs recorded: 201

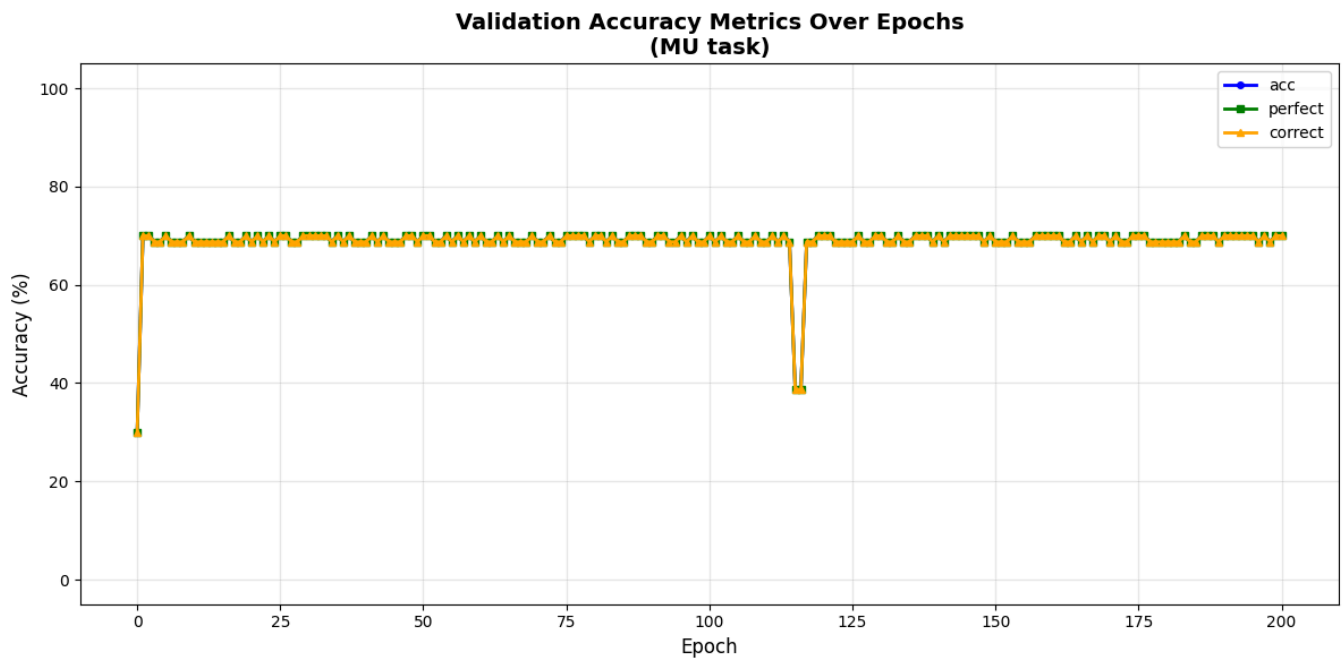
First few epochs:

	epoch	valid_arithmetic_xe_loss	valid_arithmetic_acc	valid_arithmetic_perfect	valid_arithmeti
0	0	1.128563	30.03	30.03	
1	1	0.438123	69.97	69.97	
2	2	0.427301	69.97	69.97	
3	3	0.431909	68.73	68.73	
4	4	0.434626	68.73	68.73	
5	5	0.429042	69.97	69.97	
6	6	0.447196	68.73	68.73	
7	7	0.425965	68.73	68.73	
8	8	0.427849	68.73	68.73	
9	9	0.425200	69.97	69.97	

Plot 1: Cross-Entropy Loss

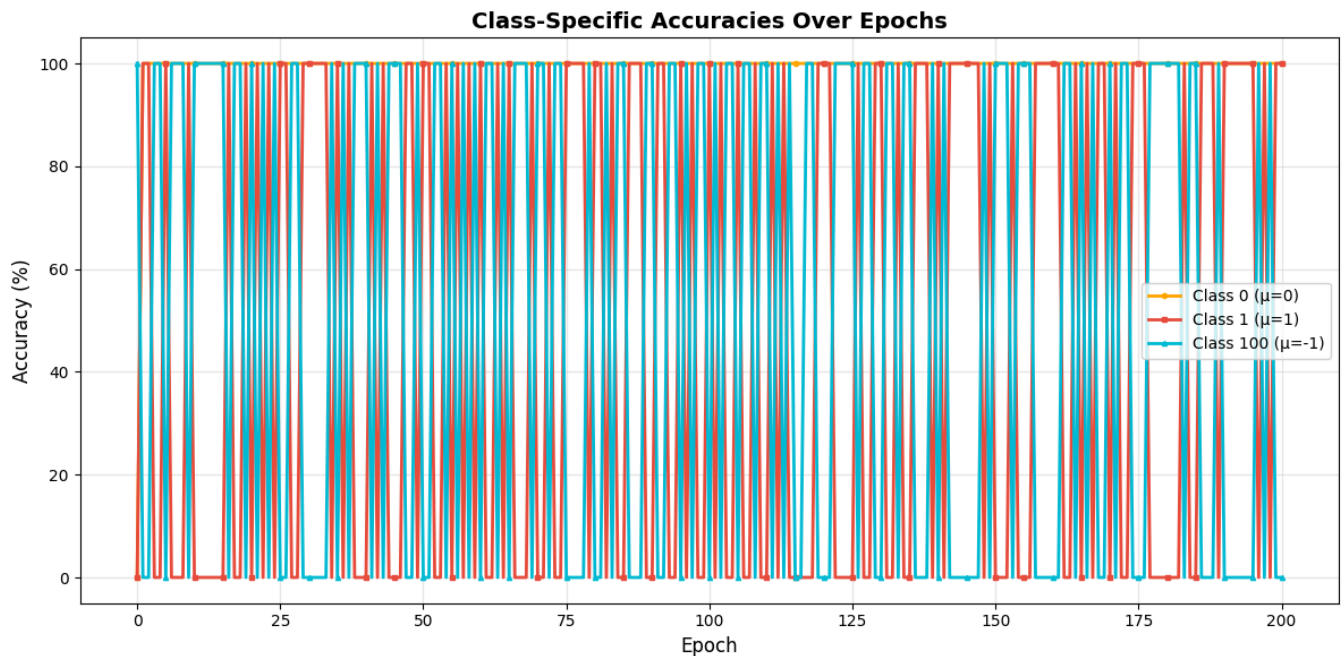


Plot 2: Accuracy Metrics (acc, perfect, correct)



Plot 3: Class-specific Accuracies (acc_0, acc_1, acc_100)

Plot saved to: ../train_log_results/model_interCRT_sqfree_natural/plot3_class_specific_accuracies.png



Best Metrics Report

BEST METRICS ACROSS ALL EPOCHS

CROSS-ENTROPY LOSS

Best Loss: 0.424774

Achieved at Epoch: 168

ACCURACY METRICS

ACC:

Best: 69.97%

Achieved at Epoch: 1

PERFECT:

Best: 69.97%

Achieved at Epoch: 1

CORRECT:

Best: 69.97%

Achieved at Epoch: 1

CLASS-SPECIFIC ACCURACIES

Class 0 ($\mu=0$):

Best: 100.00%

Achieved at Epoch: 1

Class 1 ($\mu=1$):

Best: 100.00%

Achieved at Epoch: 1

Class 100 ($\mu=-1$):

Best: 100.00%

Achieved at Epoch: 0

LATEST EPOCH METRICS:

Epoch: 200.0

XE Loss: 0.424781

Accuracy: 69.97%

Perfect: 69.97%
